HUMANOIDS2026
IEEE-RAS International Conference on Humanoid Robots

Interest Form

Loco-Manipulation Challenge

2026 IEEE Humanoids Conference, Santa Clara, California, US
December 6th-9th

The 2026 IEEE Humanoids Conference will host a competition to evaluate the capabilities and live performance of humanoid robots on standardizable locomotion and manipulation tasks. Humanoid robots have the potential to revolutionize how automation is deployed in human environments, but only if their capabilities are robust, reliable, and flexible to new and changing environments. Every year we see more impressive capabilities demonstrated by academics and companies, but still lack comparable benchmarks for performance of these robots on representative tasks. This competition is composed of baseline tasks that begin to establish a common understanding of what humanoid robots are capable of.

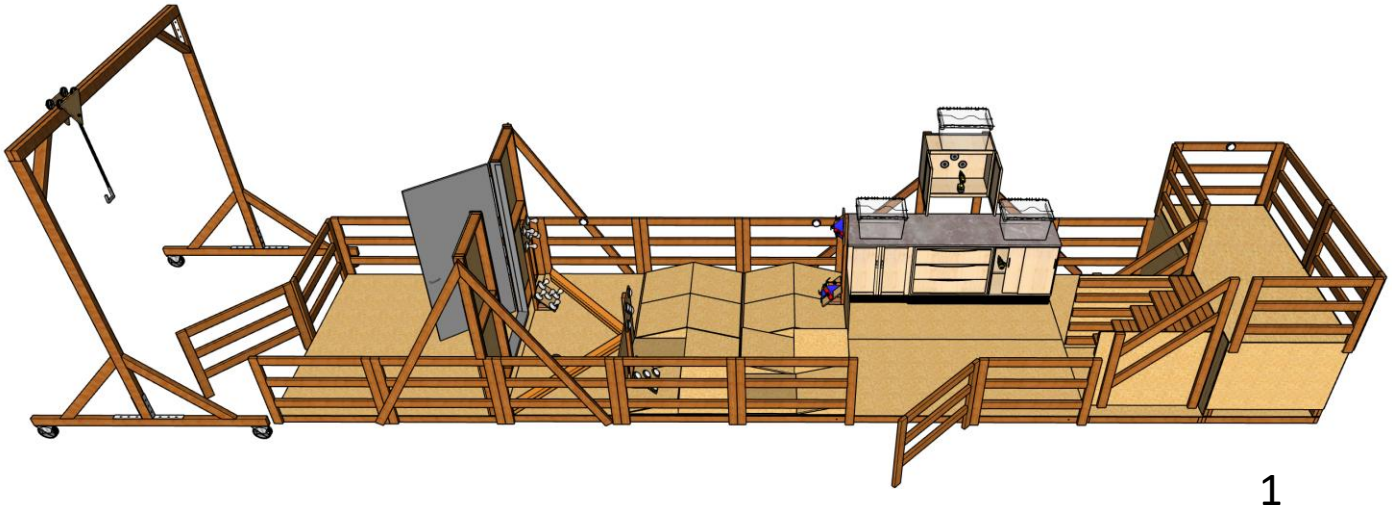
The competition is intended to be accessible to robots of any height, morphology, and autonomy level, though they should be of a general humanoid form (if you consider it a humanoid, we'll consider it a humanoid). The competition will consist of a set of mobility tasks, a set of manipulation tasks, and a combination of both in the loco-manipulation challenge. The competition is open to both university teams and companies who wish to demonstrate robot performance (though they will be judged in two different leagues). Whether the robot has been walking for years, or this is the first step out of the laboratory, teams will be able to choose which tasks they wish to accomplish, and any task completed will be scored and allow you to compete with other teams.

It is a long process to go from initial, shaky steps to hardened and reliable robots for deployment in real-world environments. This competition sets a baseline of performance for the current state of the art for humanoid robot systems, and future iterations will continue to drive the technology toward reliable, practical capabilities ready for deployment.

Baseline Humanoid Robot Capabilities

The Loco-Manipulation Challenge will consist of a single testing apparatus that contains a number of locomotion/mobility tasks and manipulation tasks. The arena includes a variety of reproducible terrains, obstacles, and manipulation tasks with varying difficulty to challenge all levels of robot performance.

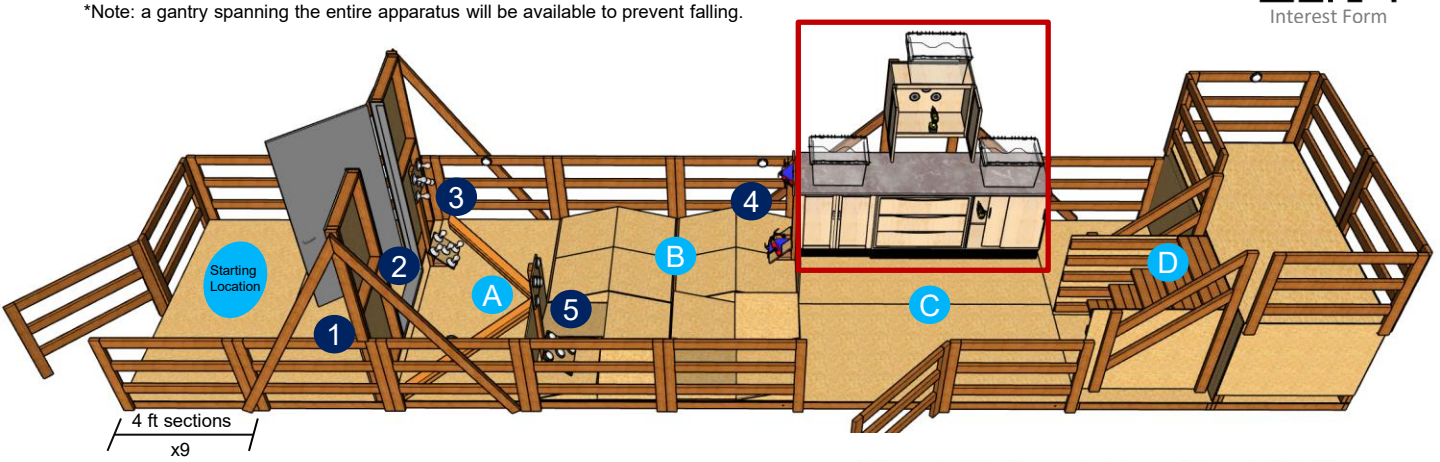
- ***Mobility*** refers to a robot's ability to walk reliably and effectively through its environment. This includes stepping over simple obstacles, maintaining stability over rough terrain, and positioning itself where tasks need to be performed. The reproducible tests measure how well a robot can get from place to place under real-world conditions.
- ***Manipulation*** refers to a robot's ability to manipulate objects with precision, control, and flexibility. While this competition does not yet evaluate for fine dexterity or force control, the kinematic interaction tasks will evaluate the robots ability to interact with general, every-day objects in a repeatable manner.
- ***Autonomy*** is designed to reduce the need for an operator to intervene in the robot's activities, either to give instruction or to correct a mistake. Autonomous operation is key to providing practical value to users of these robots. The competition will allow for any level of autonomy in operation. The lowest level of autonomy is direct teleoperation, receiving base points, but all higher levels of autonomy will be rewarded in the score.



Competition Overview 2026

Interest Form

*Note: a gantry spanning the entire apparatus will be available to prevent falling.



Manipulation with Visual Input Task:

1. **Keypad:** View a 5 digit code on a screen and press buttons according to those digits.

Manipulation tasks:

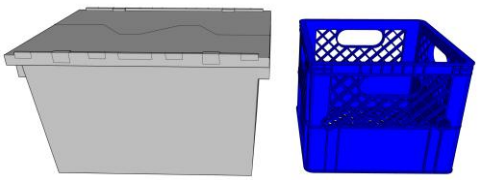
2. **Door Traversal:** Open, traverse, and safely close a door (7ftx3ft). Handle will be standard lever handle.
3. **Lightbulb Turning:** Using 3d-printed lightbulbs, tightened already placed lightbulbs and place bulbs retrieved from a box
4. **Valve Turning:** Turning of simple lever-handle water shutoff ball valves.
5. **Button Press:** Press a series of push buttons.

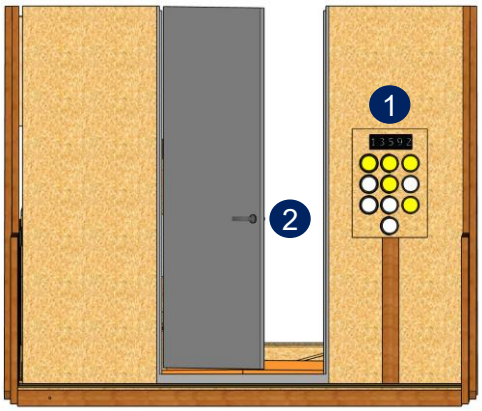
Mobility Tasks:

- **A.** **Stepover Obstacles:** A length of 4x4 lumber laid across the walkway as an obstacle.
- **B.** **Crossing Ramps:** 15° slopes in a hill and pinwheel layout.
- **C.** **Flat Ground Walking:** 8ft length.
- **D.** **Stairs Ascend and Descend:** Walking up and back down a half-flight of stairs.

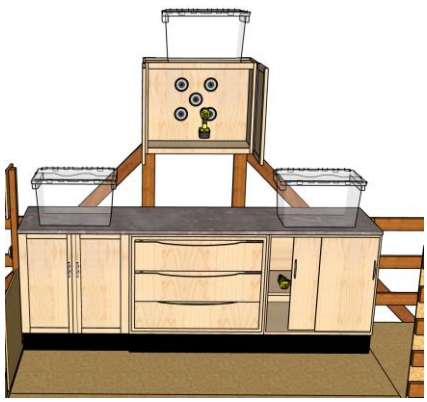
Loco-Manipulation:

- **Simultaneous Performance:** Perform all manipulation and mobility tasks simultaneously.
- **Bonus: Tote Carrying** Perform the mobility tasks while carrying a tote or milk crate.





View from the starting location.



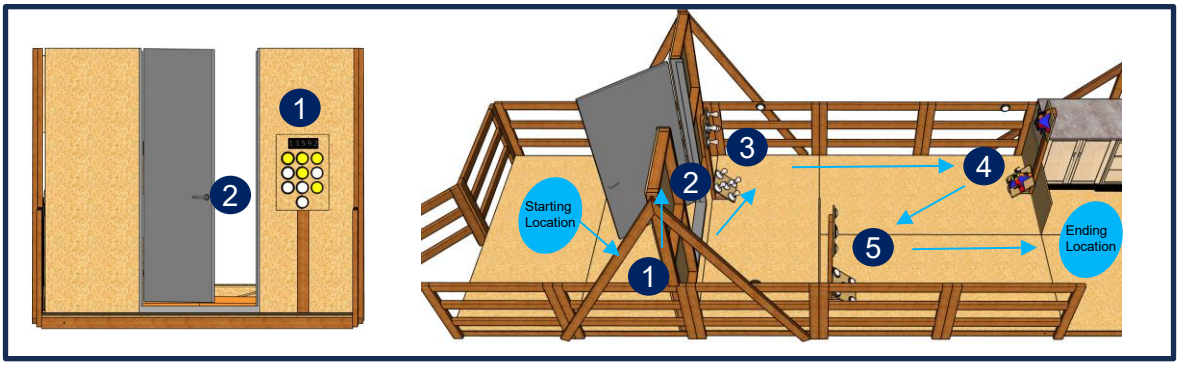
Demonstration Tasks: These tasks include

- Tote moving – Lateral and Vertical motion
- Drawer and cabinet opening
- Loose object pick and place
- Hand tool Manipulation.

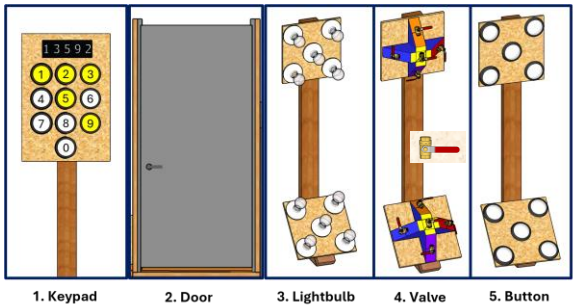
These tasks will not be scored and are present only for demonstration of future evaluation tasks. They may be attempted if teams wish.

Competition Task Details

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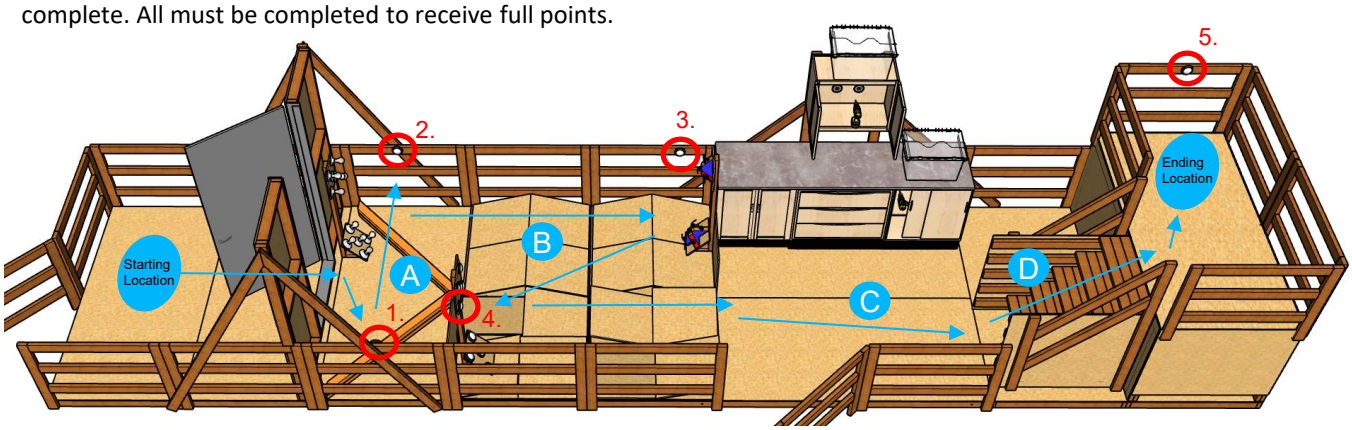


Manipulation Tasks: during Manipulation Trials the obstacles and rough terrain will be removed from the test apparatus so a robot only needs to perform flat ground walking to walk from the starting location, to each manipulation task being attempted in order, and then to the ending location (also compatible for wheeled humanoids). Once complete, the robot will repeat the tasks in reverse order, returning to the starting location. This will constitute one cycle. This cycle can be repeated until time is up.



1. **Keypad:** A code will be displayed and the robot must press the corresponding buttons, in order. The buttons will light up when pressed sufficiently.
2. **Door Traversal:** The door is 3ft x 7ft. The handle must be turned, door pushed open, and held open as the robot walks through the doorway. The door hinges will have torsional springs to cause the door to automatically close.
3. **Lightbulb:** Commonly available light fixtures will be affixed to a board. 3d-printed lightbulbs (no risk of breaking glass) must be either just tightened or picked up and screwed into place.
4. **Valve Turning:** Commonly available 90° lever handle ball valves will be mounted in multiple orientations. They must be turned from a perpendicular to a parallel orientation.
5. **Button Pressing:** The same buttons as on the keypad will be mounted on a board and must simply be pressed.

The lightbulb, valve, and button tasks will be mounted on the wall and at the floor, requiring the robot to bend over to complete. All must be completed to receive full points.



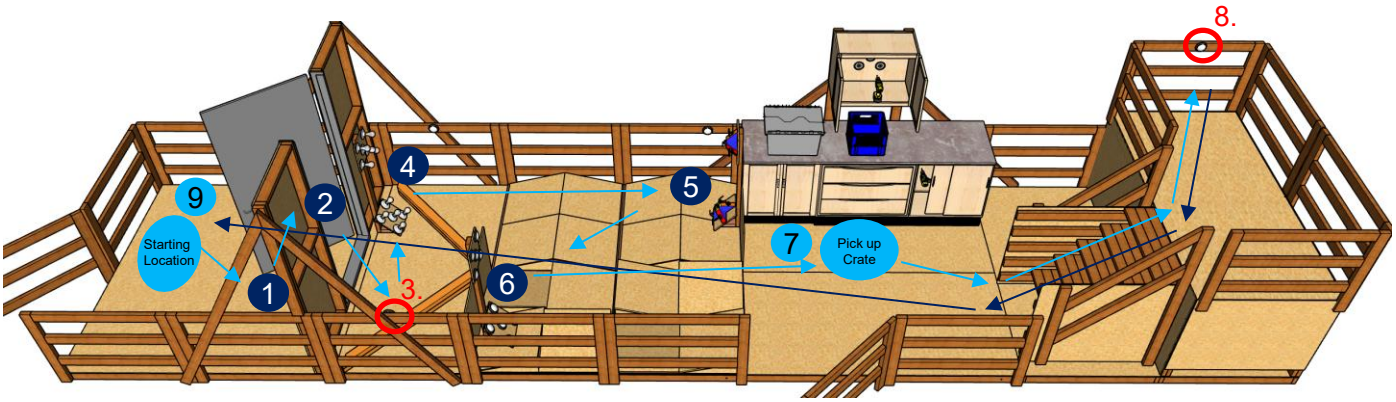
Mobility Tasks: during Mobility Trials the obstacles and rough terrain will be replaced. There will be a series of four tasks. To mark completion of motion in a certain direction, there are also a series of 5 buttons (circled above) that must be pressed. This signifies reaching a key location:

A. **Stepover Obstacles:** From the starting location a robot will walk through the open doorway and then turn right to press the first button, turn around and proceed to press the second. To do these motions the robot will step over two lengths of 4x4 lumber laid across the walkway as an obstacle.
B. **Crossing Ramps:** The is an 8ft x 8ft section of terrain that consists of 15° slopes in a hill and pinwheel layout. The robot will proceed over this to press the two buttons located at the two manipulation tasks.
C. **Flat Ground Walking:** The robot will walk the 8 foot flat ground section length to the bottom of the stairs.
D. **Stairs Ascend and Descend:** The robot will walk up the stairs to the top landing, turn left and press the last button.

The robot will repeat the above tasks in reverse order, returning to the starting location. This will constitute one cycle. This cycle can be repeated until time is up, scoring points for the total number of tasks completed.

Competition Task Details

Interest Form

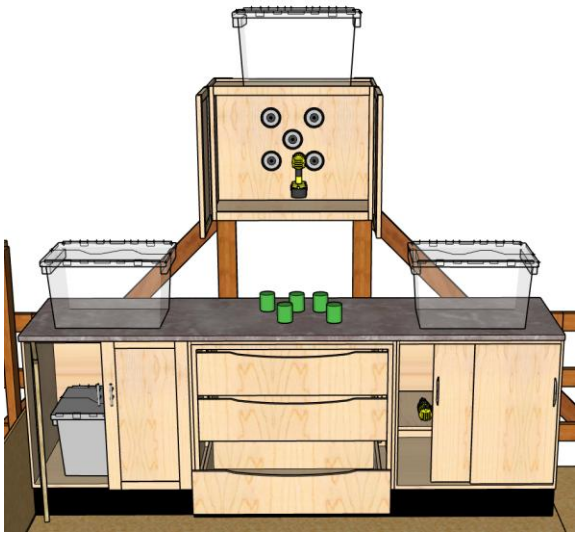


Loco-Manipulation Tasks: The third phase of the competition will be simultaneous locomotion and Manipulation Tasks. The obstacles and rough terrain will be included. Beginning in the same starting position, the robot will **1).** Complete the keypad task **2).** Open and traverse the door **3).** Turn right and press the first button, then **4).** perform the lightbulb, **5).**Valve turning, and **6).** Button pressing tasks in that order. **7).** The robot will then pick up one of the two crates from the counter, **8).** Walk up the stairs to press the button on the landing, and finally **9).** Descend the stairs and walk back through the terrain to the starting position, all while carrying the crate, and place the crate on the ground. Completing all of these tasks will be one cycle, and this cycle can be repeated until time is up. (One of the staff will reset the crate from the starting location to the counter.

Task Choice and Overall Performance: During the above three parts of the competition, teams can choose which tasks they wish to attempt. Teams can skip tasks they do not have confidence in completing, or can attempt a task, fail, and choose to move on anyway. Points will be scored for every task completed, and extra points will be scored for performing all tasks as laid out above (ie. Extra points will be missed if any tasks are skipped). Teams can choose to not participate in any parts of the competition, meaning if a team only wants to try mobility tasks and no manipulation, or vice versa, they may. Awards will be given for each part: Manipulation, Mobility, and Loco-Manipulation. An overall winner will also be identified from the aggregate score in all parts of the competition.

Demonstration Tasks: Included in the competition apparatus is a section that contains tasks that **will not be a scored part of the competition.** These tasks are more complex tasks that require constrained motion, tool manipulation, higher range of motion actions and more. These tasks are available for demonstration purposes. The competition apparatus is the initial version of a larger testbed that is being designed for all industry, meaning companies with the best performing robots in the world will use the testbed to evaluate their robots. Tasks like these will be included. Additionally, **these tasks will be included in future years of this competition.** So teams can attempt the tasks now, but have at least another year to properly learn to perform the tasks for competition.

1. **Tote Moving:** A tote or milk crate will be moved to varying locations, including within a lower cabinet (that must be opened by the robot) and on top of the upper cabinet.
2. **Drawer Opening and Object Collection:** Various objects (simply represented by green cylinders here) will be on the counter, and must be placed in the drawers.
3. **Hand Tool Manipulation:** In the lower-right hand cabinet is a common hand drill. The cabinet must be opened, drill picked up, then manipulated to touch the targets located in the upper cabinet with the included drill bit.



Competition Features

Interest Form

Competition Schedule: The Loco-Manipulation Challenge is planned to be run over the course of two full days during the 2026 IEEE Humanoids Conference. There will be a preparation day in which teams will have access to the competition setup prior to the beginning of evaluation trials. There will be 3 parts to the competition, the manipulation tasks, the mobility tasks, and the simultaneous mobility and manipulation tasks, called the Loco-Manipulation. These will be held in series meaning that all participating teams will compete on the Manipulation tasks, then on the Mobility tasks, and then finally on the Loco-Manipulation, in that order, over the course of 2 days.

Task Choice: The competition is meant to be accessible to all robots. This means that a robot does not need to be capable of all manipulation and mobility tasks to compete. Each team will choose which parts of the competition they wish to participate in, meaning teams can choose to only participate in the manipulation or only in the mobility parts of the competition if they wish. There will be awards for the best at manipulation, the best at mobility, and then the best overall performing robot across all 3 parts.

Trial Time Limits: All competition trials will be timed. Each team will have around 10-20 minutes (depending on number of teams and scheduling) to perform the manipulation, mobility, or loco-manipulation tasks, depending on which is being attempted at that time. The goal will be to complete all of the tasks as many times as they can within that time. The time limit is not intended to make it a race. Rather, there is enough time for a capable robot to demonstrate a statistically significant number of task repetitions. This provides a measure of reliability that the task can be performed. There will be setup and exit time before and after each trial.

Scoring: Within each part of the competition there are individual tasks to complete. Points will be scored for every task completed, and completion of each task is not required. This means that if a robot is only capable of performing 2 out of the 5 manipulation tasks, it can perform just those tasks. Extra points will be scored for completing all tasks available

Autonomy: Any level of autonomous operation can be implemented for this competition (e.g: teleoperation, commands from an operator computer, or full autonomous operation). This level of autonomy, except for full autonomy, can change depending on the task being performed. The level of autonomy should be declared during operation and more autonomy will receive a higher score. Full autonomy requires receiving no input from a remote operator from start to end time. Pre-mapping of the test apparatus will be allowed on the preparation day to support autonomous operation.

Robot: Each team can register **one robot to compete**. Robots must weigh under **120 kg**. Robot must be able to fit through a 3ft wide doorway to navigate the apparatus. All robots must be equipped with secure, safe handles to move the robot, as well as a rope/cable attachment point to attach the overhead gantry. All robots must have an e-stop that is to be manned at all times during operation. The robot should operate with battery power while competing.

Remote Operator Stations A team member will be designated as the remote operator, who will control the robot during while competing, if necessary. A specified station for the remote operator will be provided (table and chair) with a clear line-of-sight to the robot within the test apparatus. No communication between team members and the operator is allowed unless a failure occurs. The operator must receive all situational awareness through the system interface as if the robot is down range or in a structure. **One operator at a time only! Switching operators is OK. Operators must stay at the operator station and must not watch the robot during the trial.**

Failures: An operational failure occurs when the robot is rendered inoperable. If this occurs, a penalty in scoring will be applied, but the team will be allowed to recover and reset the robot. Time will not be halted during this resetting. After reset, the tasks can be resumed. Failure not requiring intervention (e.g falling over but managing to self-right, or automatically rebooting if necessary) will receive a lesser penalty, and no pause is needed. Engaging the gantry to support the robot will be considered a balance failure.

Tethers: Due to potential wireless traffic interference at the conference venue, the robot may be communicated with via a cable, if necessary. To avoid getting snagged on obstacles, cable tethers should be spooled on the robot and act as a winch when necessary to help descend stairs then climb back up if necessary. If tethers are dragged behind the robot within the test lanes, the tether must be managed from the lane entrance by a designated cable handler. Tethers may not be guided over the lane walls. **Only one cable handler is permitted during the mission. The cable handler must remain at the lane entrance, and the use of sticks or similar tools to aid in cable management is prohibited.**

Personal Safety: Teams must be dressed appropriately to enter the arena area; closed toed shoes and long pants.

Participation and Next Steps

Interest Form

Participation: Teams from all Universities are invited to participate. Humanoid robot manufacturers are also invited to participate. These two types of teams will be separated into two different leagues for scoring and awards. All teams interested in participating should fill out the **Team Participation Interest Form, linked here and available at the top-right QR code**. This is the first step to expressing interest in participation in the competition.

Team Description Paper (TDP): The next step for teams wishing to participate in the competition will be the submission of a brief Team Description Paper in November prior to the competition. This paper will require sharing the details of the physical robot, control and autonomy capabilities, and other features of the robot. It will require evidence of conformity to the rules of the competition, basic operating capability of the robot, and final details on the attending team. **The complete and final rules for the competition as well as a template for this TDP will be made available during Fall 26.** The TDP will not be scored as part of the competition. The TDP will be reviewed for basic qualification to participate in the competition. If there are too many teams intending to participate (more than can be accommodated in the space and time available at the conference venue) final selection of teams will be done by quality of TDP and demonstrated likelihood to perform during the competition. Submission of the TDP is how a team formally applies to participate in the competition.

Travel Awards: There is a potential for travel awards being made available to participating teams to aid in covering travel expenses. These awards are not guaranteed, but if available will be distributed as fairly as possible to teams who submit a TDP.

Further Resources and Documents: This document you are reading is the competition description. The following documents will be released prior to the competition:

- **Competition Apparatus Plans and Models:** in this document are pictures and details of the physical setup of the apparatus for the competition. This apparatus is built from materials available at any hardware store. Once ready, full 3d models of the apparatus, including a bill of materials and build guide will be made available. Virtual representations for loading into common simulation softwares will also be provided.
- **Final Rules:** this will include the exact task details, competition procedure, full competition layout, and complete scoring information.
- **Team Description Paper Template:** A full template doc will be released that teams can use to easily prepare the TDP submission, required for participation in the competition.

These documents will be released during Fall 2026. All additional details will be released on the competition page of the IEEE conference website. All who fill out the Team Participation Interest Form will be notified via email of these details when they are made available.

Organizers: The Loco-Manipulation Challenge has been designed and is being organized by the National Institute of Standards and Technology (NIST), and specifically staff from the Measurement Science for Robotics and Autonomous Systems (MSRAS) Program. NIST staff designed the competition tasks for the DARPA Robotics Challenge (2013-2014), and currently run other competitions such as the IEEE/RAS Legged Robot Challenge at ICRA (2023-2026), the RoboCup Rescue International competition (2001-2026), and more. More information on the industry benchmark this competition is in support of can be found here: NIST Humanoid Robot Baseline Benchmark, currently under development.

Continuing Apparatus Design: Note that the competition apparatus shown in this document may have some small changes from this to the final version. The tasks themselves will not change, but some dimensions, structural components, and other details will. The final version will be noted as such once released.

Additional Questions: If you have any additional questions about the competition, please contact the conference competition chairs at humanoids2026-cc@ieee.org.